

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	N. PARNITA	Reg. No.:	192525322
Section / Batch:	2025-2029	Date:	3.8.26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

- 1) Analyze the performance of Linear Search for unsorted attendance.
- 2) Explain Brute Force String Matching, Analyze it.

Code of Conduct

I, N. PARNITA (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

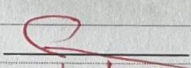
Signature of the Student: N. Parnita

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario.
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				
Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: 	Signature: _____	Signature: 
Date: 3/8/20	Date: _____	Date: 3/8/20

- 1) A University system stores attendance records unsorted due to frequent updates. Faculty need to quickly locate specific student records.

1) How Linear Search Works

Linear search checks each attendance record one by one from the beginning until the required student is found or all records have been checked. It is suitable because the records are unsorted.

2) Performance Analysis

- * Best Case : $O(1)$
- * Average Case : $O(n)$
- * Worst Case : $O(n)$

3) Would Sorting Improve Efficiency?

Yes, If the records are sorted, Binary Search can be used, reducing the search time from $O(n)$ to $O(\log n)$. However, since attendance records are updated frequently, maintaining a sorted list requires additional sorting time.

4) Conclusion

Brute Force String Matching is simple but inefficient for large texts. Optimized Algorithms provide faster searching.

START



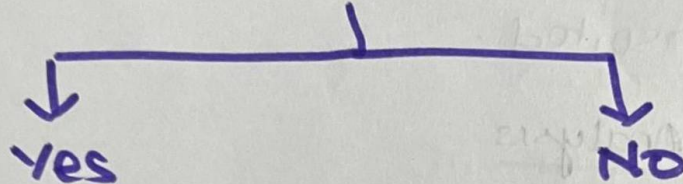
Receive Student ID to Search



Start from First Record



Compare Record with Student ID



Record Found

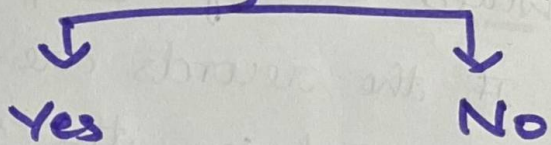


Display Record

Move to next Record



More Records?



→ Repeat

Record not Found



END

2) A code editor searches for variables and keywords in large programs using Brute-Force String Matching.

1) How Brute Force String Matching Works

The algorithm compares the search pattern with text character by character. If a mismatch occurs, it shifts the pattern by one position and repeats until the pattern is found or the text ends.

2) Time Complexity

* Best case : $O(n)$

* Average case : $O(n \times m)$

* Worst case : $O(n \times m)$

Where n = length of text, m = length of pattern

3) Why Optimized String Searching is Needed

For large programs, Brute-Force String Matching performs many repeated comparisons, making it slow. Optimized algorithms like KMP and Boyer-Moore reduce unnecessary comparisons, making searches faster, and more efficient.

4) Conclusion :

Brute Force String Matching is simple to implement but is not efficient for large text files. Algorithms provide better performance & searching.

